

CN Unit 4 — Transport Layer (TCP/UDP) Complete Notes

Computer Networks (CN) · Unit 4 · Transport Layer (TCP/UDP) · 2025

Overview

These notes cover **Transport Layer (TCP/UDP)**. From bits on the wire to HTTP — layered network architectures, routing, transport protocols and the applications on top. Each section states the core idea, a worked intuition and the points examiners look for.

1. UDP

UDP is a connectionless, best-effort transport with minimal header overhead.

Why it matters: UDP appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of udp in one line.
- A small example showing udp in practice.
- Common mistakes students make and how to avoid them.

2. TCP Connection Management

TCP opens connections with a three-way handshake and closes them with a four-way exchange.

Why it matters: TCP Connection Management appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of tcp connection management in one line.
- A small example showing tcp connection management in practice.
- Common mistakes students make and how to avoid them.

3. TCP Flow Control

The receive window prevents a fast sender from overwhelming a slow receiver.

Why it matters: TCP Flow Control appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of tcp flow control in one line.
- A small example showing tcp flow control in practice.
- Common mistakes students make and how to avoid them.

4. TCP Congestion Control

Slow start, congestion avoidance, fast retransmit and fast recovery adapt the congestion window.

Why it matters: TCP Congestion Control appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of tcp congestion control in one line.
- A small example showing tcp congestion control in practice.
- Common mistakes students make and how to avoid them.

5. Port Numbers and Sockets

Ports multiplex transport connections; sockets are endpoints identified by IP and port.

Why it matters: Port Numbers and Sockets appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of port numbers and sockets in one line.
- A small example showing port numbers and sockets in practice.
- Common mistakes students make and how to avoid them.

6. Quality of Service

Leaky bucket and token bucket algorithms shape traffic to meet QoS goals.

Why it matters: Quality of Service appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of quality of service in one line.
- A small example showing quality of service in practice.
- Common mistakes students make and how to avoid them.

Quick revision

UDP · TCP Connection Management · TCP Flow Control · TCP Congestion Control · Port Numbers and Sockets · Quality of Service